

INNOVATIONS

1 Electronic nose (e-nose) for aroma characterisation of black tea

An emergent need for some physical, fast and low-cost methodology for measurement of physical quality attributes like smell has been felt by the food processing industries, especially black tea manufacturing industries in India. The e-nose system proposes a pragmatic methodology for measuring the aroma of black tea by fusion of advanced sensor technologies backed up with intelligent self-learning-type software algorithms.

The e-nose-based aroma indices innovated through sustained experimentations in black tea fermentation process enable online smell monitoring and detection of optimal fermentation time for producing best-flavour components in the manufactured black tea. The neural network-based self-learning computational models built into the e-nose software system permit fast declaration

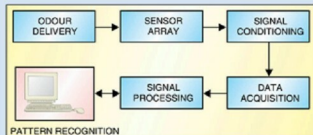


Fig. 2: Block diagram of e-nose system

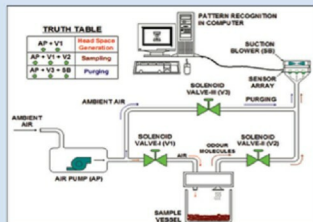


Fig. 3: E-nose set-up

of quality scores of finished black tea, coherent with the organoleptic appraisals of the tea-tasters. The entire system is software-driven, field-customisable and requires minimal operator intervention.

E-nose detects and discriminates among complex odours using a sensor array. The sensor array consists of non-specific sensors that are treated with a variety of chemical materials. The sensor array is exposed to the volatile molecules and smellprint (or fingerprint) is generated from it. Patterns from known odours are used to construct the database and train a pattern-



Fig. 4: Software screen for e-nose

recognition system so that unknown odours can be classified and identified.

Fig. 2 shows the block diagram of e-nose system. The e-nose instrument comprises three elements—sample handling system, detection system and data processing system.

Sample handling is a critical step affecting the analysis by e-nose. To introduce the volatile compounds present in the head space (HS) of the sample into the detection system of e-nose, several sampling techniques are used:

1. Static head space (SHS) technique
2. Purge and trap technique
3. Solid-phase micro extraction technique
4. Stir bar sorptive extraction technique
5. Inside-needle dynamic extraction technique
6. Membrane introduction mass spectrometry technique

Out of these, SHS technique has been observed to be useful for food and agro applications.

The most complicated part of electronic olfaction process is odour-capture and the sensor technology deployed for it. Any sensor that responds reversibly to a chemical in gaseous or vapour phase has the potential to be developed in an e-nose format. Early e-nose used either polished wires in contact with porous rods saturated with different electrolytes, or thermistors coated with materials such as gelatin, fats or polymers. In the 1980s, advances were made with the appearance of chemically-sensitive sensors and developments in electronics and computing.

Some of the essential or desirable properties of the chemical micro-sensors to be used in e-nose are selectivity, sensitivity, speed of response, reproducibility, reversibility and portability.

Table III shows sensor types and associated measuring principles used for e-nose. Out of these sensors, conducting polymer, metal-oxide semiconductor and bulk acoustic devices are most commonly used in commercial e-noses.

Advanced signal processing. In a machine olfaction system, two major building blocks are the sensors and the pattern-classification engine. Interfacing electronics, signal conditioning and data acquisition circuits provide the crucial link between these two blocks.